Here's the detailed summary of *Le code de l'immortalité* (Radman, Humensciences, 2019), emphasizing the science.

What's new versus the 2011 book

This book is half scientific argument, half memoir (his Dalmatian childhood, the war, his path from Split through Brussels, Paris, Harvard, and back to a research institute in Split). The scientific core sharpens and extends the 2011 thesis, with two genuinely new pieces: **cellular parabiosis** as a tissue-level mechanism, and the move from concept to a concrete **skin product based on "chaperone molecules."** His framing slogan: the project is "the battle for health, not the war against diseases."

The unified thesis: one cause, many diseases

Radman doubles down on the claim that aging and degenerative disease share a single upstream cause — **protein oxidation (carbonylation)**, his "biological rust." His strongest supporting datum: the *kinetics* of protein oxidation are the same across species — the accumulation curve over a nematode's three-week life is identical in shape to a human's over a hundred years. Protein oxidation is therefore the universal biomarker of aging. From this he argues there is not one process per disease plus another for aging, but a single process producing cancer, Alzheimer's, diabetes, and cardiovascular and neurodegenerative disease alike.

A key conceptual inversion he stresses here: oxidation damage to proteins is *more* mutagenic for DNA than direct damage to DNA. If DNA is damaged but the repair proteins are intact, they fix it; if the repair proteins are oxidized, DNA gets hit hard. So proteins are upstream of DNA — the reverse of the prevailing DNA-centric dogma. After forty years working on DNA, he reframes DNA as consequence rather than cause.

Deinococcus radiodurans and the protective metabolites

The corned-beef discovery story recurs, but the mechanistic account is tighter. Two findings:

DNA reconstitution. The bacteria keep multiple genome copies (at least two even when dormant), and reassemble shattered DNA by overlap-matching fragments across copies — slowly but error-free, sometimes over days.

The protective "something extra." With Anita Kriško, his main collaborator for the decade, he ultrafiltered *Deinococcus* extract through a molecular sieve passing only molecules under ~3,000 daltons (proteins run 100,000–200,000 Da). This isolated small **metabolites**. When *E. coli* and human proteins were placed in this ultrafiltered "juice" and irradiated, they were protected. The metabolites act as **sacrificial antioxidants** — they oxidize in place of the far more precious proteins, neutralizing free radicals before oxygen atoms fix onto the proteins' carbon atoms. His analogy: specialty steels that oxidize at the surface to keep rust from reaching the core. Crucially, the protection works identically on bacterial and human proteins.

Cellular parabiosis — the new mechanism

This is the headline new idea. Cells in tissue aren't isolated; they exchange macromolecules (including proteins) through nano-channels he calls **TNTs (Tunneling Nanotubes)**, whose existence was shown in the late 1960s. Through this exchange, healthy cells supply good-quality components to a weaker neighbor whose own version of some protein is defective (from a destabilizing silent mutation). The defect isn't cured, but its consequences are buffered — mutual compensation creates **organ-level robustness**. He cites hepatocytes (~three-quarters of liver cells) effectively functioning as one giant cell, explaining the liver's regenerative capacity.

The inflammation link is the mechanistic payoff. During infection, large macrophages release **metalloproteases** to push through the extracellular space, degrading the proteins that form the nanotubes. This temporarily isolates cells in the "battlefield" zone. Acute, violent inflammation (a fever spike) is actually beneficial — it clears extracellular clumps of dead, oxidized protein. But **chronic low-grade inflammation** is the destroyer: oxidized protein aggregates resist degradation, the immune system keeps attacking them and tearing down surrounding nanotubes, and cells cut off from their neighbors degenerate in whole groups. Hence cancer and Alzheimer's are almost always tied to chronic inflammation.

The predictive nematode experiment

A clean demonstration that oxidation is cause, not consequence. Self-fertilizing hermaphrodite nematodes produce isogenic, isochronic clones — same genome, same birthday, and postmitotic (968 non-renewing cells, so no replication error or mutation can confound things). Aligned on conductive agar and nudged forward by a weak current, they sort into fast, medium, and slow movers. Measuring protein carbonylation versus survival curves shows the slow movers oxidize most and die youngest — and oxidation level at day two predicts individual lifespan. Since nothing genetic differs, the oxidation rate *itself* sets longevity. This grounds his ambition to measure biological (versus chronological) age and predict disease predisposition from birth.

Why big animals don't get proportionally more cancer

He brings in **Peto's paradox** (via his friend Richard Peto): humans have ~1,000× more cancer-risk somatic stem cells than mice, with similar per-cell mutation rates, yet don't get 1,000× more cancers. Berenblum's mid-1940s work on cancer latency offers the lead: a cancer's latency runs to about a third of the species' potential lifespan. A mouse cell "holds back" ~six months; a human cell, twenty-plus years. The size–longevity correlation really reflects how long an organism must stay healthy to reproduce enough to perpetuate the species. Once reproduction is "done," nature has no interest in the extra mouth — latent diseases are no longer restrained, which is why cancers and degenerative disease explode in humans after ~45.

From lab to product, and the therapeutic vision

He frames protein protection as potentially as radical as vaccination — because medicine has almost only curative treatments and essentially just two mass *preventive* ones (antibiotics and vaccines). The concrete near-term output: the first **skin product using chaperone molecules**, drawn directly from nature, about to go on sale, with further products aimed at preventing or reducing degenerative disease to follow. The therapeutic toolkit he envisions: diffusing

"chemical chaperone" molecules that shield a mutation-fragilized protein's vulnerable site, or refolding-corrector molecules that help defective proteins fold properly and resist oxidation. He invokes the disappearance of tuberculosis sanatoria as his model and dreams of cancer centers closing the same way.

The memoir and worldview threads

Interleaved: his birth in 1944 in a Split cellar under Allied bombing, near-starvation rescued by a wet-nursing aunt (his first "selection test"), the family's singing-as-tooth-cleaning ritual, his self-taught literacy; later his scientific formation and the institute he built in Split (MEDILS). Recurring philosophical positions carry over from 2011 — the primacy of prevention over therapy, criticism of fragmented, short-termist, commercially-driven cancer research that stays "at the surface," and his preference for synthesis over detail. He keeps Théodore Monod's line that utopia means not the unrealizable but the unrealized.

Bearing on your dermocosmetics work

The 2019 book is more directly relevant to NAOS than the 2011 one, because it crosses from concept to product. The exploitable threads: **carbonylation as the master biomarker**; the **<3 kDa sacrificial-antioxidant metabolites** from extremophiles acting cross-species on human proteins; **chaperone / chemical-chaperone and refolding-corrector molecules** as the protective modality; **TNT-mediated parabiosis** and **chronic-inflammation-driven nanotube destruction** as a tissue-level (skin-relevant) mechanism; and his explicit positioning of a topical skin product as the first translation — which is precisely the kind of "strategic hub" angle (protein-protection, anti-carbonylation) your client has been pushing on.

Want me to pull the parabiosis/TNT material together into a tighter technical brief, or translate a few specific passages (e.g., the metabolite-isolation method or the nematode protocol) precisely for citation?